

JEE ADVANCED WEEKEND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



Sec: SR.IIT_*CO-SC(MODEL-A&B)
Time: 3 Hrs

Date: 30-06-24
Max. Marks: 183

30-06-24_SR.IIT_STAR CO-SC(MODEL-A&B)_JEE ADV_WAT-52_SYLLABUS

- MATHEMATICS:** **PRESENT WEEK(80%):** Number of integral divisors, Complete combinations, Number of integral solutions of linear equations, Derangement
PREVIOUS WEEK(20%): **Permutations & Combinations:** Fundamental Principles of counting, Linear permutations, Circular permutations
- PHYSICS:** **PRESENT WEEK(80%):** **WAVE OPTICS:** Wave nature of light- wave fronts - Huygens principle (Application to reflection, refraction), Young's double slit experiment and variations of standard YDSE, **(EXCLDE: Diffraction due to a single slit, POLARISATION)**
PREVIOUS WEEK(20%): **OPTICS-III :** Refraction at curved surface ,Lenses and combination of lenses
MAINS EXPERIMENTS: The focal length of Convex lens, using the parallax method.
ADVANCED EXPERIMENTS: focal length of a concave mirror and a convex lens using u-v method
- CHEMISTRY:** **PRESENT WEEK(80%):** **CO-Ordination Compounds:** Introduction, Definition of important terms related to co-ordination compound, types of ligands, IUPAC rules, Nomenclature, Werner theory and Sidgwick theory, VBT -hybridization and geometries (linear, tetrahedral, square planar and octahedral)
PREVIOUS WEEK(20%): Colour, Magnetic properties, Catalytic properties, alloy and interstitial compounds formation, oxides and oxoanions of metals, Preparation & Properties of Potassium dichromate, F-BLOCK

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 8 – 12)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 13 – 18)	Three column paragraph Questions With Single Answer Type	+3	-1	6	18
Total				18	61

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 25)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 26 – 30)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 31 – 36)	Three column paragraph Questions With Single Answer Type	+3	-1	6	18
Total				18	61

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 43)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 44 – 48)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 49 – 54)	Three column paragraph Questions With Single Answer Type	+3	-1	6	28
Total				18	61

THE NARAYANA GROUP

SECTION – I

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

1. The coefficient of x^{50} in the expansion of $\sum_{k=0}^{100} C_k (x-2)^{100-k} 3^k$ is also equal to :
 - A) Number of ways in which 50 identical books can be distributed in 100 students, if each student can get atmost one book.
 - B) Number of ways in which 100 different white balls and 50 identical red balls can be arranged in a circle, if no two red balls are together.
 - C) Number of dissimilar terms in $(x_1 + x_2 + x_3 + \dots + x_{50})^{51}$.
 - D) $\frac{2 \cdot 6 \cdot 10 \cdot 14 \dots 198}{50!}$
2. If 'K' is the number of ways in which we can choose 5 letters from the word INTERNATIONAL then-
 - A) K is a three digit number
 - B) K is divisible by 4
 - C) K is divisible by 11
 - D) K is divisible by 9
3. The number of words which can be made from letters of the word INTERMEDIATE is
 - A) 907200 if words start with I and end with E
 - B) 21600 if vowels and consonants occupy their original places
 - C) 43200 if vowels and consonants occur alternatively
 - D) 302400 if all the vowels occur together.
4. If $p = 2^a \cdot 3^b \cdot 5^c, a, b, c \in N$, is the number of words which can be formed using all the letters of the word 'NAMAKEEN' so that no two alike vowels are together then :
 - A) number of odd divisors of p is 6 .
 - B) number of even divisors of p is 42 .
 - C) number of zeroes at the end of p is 2 .
 - D) number of zeroes at the end of p is 1 .

5. Number of ways in which 32 identical blankets can be distributed among 4 persons. So that each of them gets number of blankets which are divisible by 2 but not by 4, is
 A) 9C_3 B) ${}^{19}C_3$
 C) 9C_6 D) ${}^4C_1 + 5 \cdot {}^4C_2 + 10 \cdot {}^4C_3 + 10 \cdot {}^4C_4$
6. If the total number of integer n such that $2 \leq n \leq 2014$ and H.C.F of n and 36 is one is N , then which of the following is true ?
 A) N is a 3-digit number
 B) N is Prime
 C) Number of positive integral divisors of N is 8
 D) Sum of digits of N is 13
7. Let a, b, c, d be non zero distinct digits. The number of 4 digit numbers $abcd$ such that $ab + cd$ is even is divisible by:
 A) 3 B) 4 C) 7 D) 11

SECTION-II
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.

8. If the possible number of positive integral ordered triads (m, n, p) where $m, n, p \in N$ such that $1 \leq m \leq 100, 1 \leq n \leq 50, 1 \leq p \leq 25$ and $2^m + 2^n + 2^p$ is divisible by 3 are K , then the value of $\frac{K}{6250}$ is _____
9. Let N be the number of integral solution of the equation $x + y + z + w = 15$ where $x \geq 0, y > 5, z \geq 2$ and $w \geq 1$. Find the unit digit of N .
10. There are 10 stations enroute. A train has to be stopped at 3 of them. Let N be the ways in which the train can be stopped if atleast two of the stopping stations are consecutive. Find the value of $\sqrt{N}$
11. Nine people sit around a round table. The number of ways of selecting four of them such that they are not from adjacent seats, is _____
12. Let the number of arrangements of all the digits of the numbers 12345 such that atleast 3 digits will not come in it's original position is N . Then the unit digit of N is _____

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q,13, Q,14 and Q,15 by appropriately matching the information given in the three columns of the following table.

Consider all possible permutations of the letters of word 'ENDEANOEL'

Column-I contains values of m

Column-II contains values of n

Column-III contains values of $\frac{m}{n}$

	Column I values of (m)	Column II (n)	Column III $(\frac{m}{n})$
A)	The number of permutations containing the word ENDEA is	P) 5	i) 420
B)	The number of permutations in which the letter E occurs in the first and the last position is	q) 3	ii) 24
C)	The number of permutations in which none of the letters D,L,N occurs in the last five positions is	r) 6	iii) 80
D)	The number of permutations in which the letter A,E,O occur only in odd positions is	s) 4	iv) 60

13. Which of the following is CORRECT COMBINATION?

A) (A), (P),(ii) B) (C),(r), (iii) C) (D), (q) ,(i) D) (B),(s),(ii)

14. Which of the following is correct combination?

A) (B),(r), (ii) B) (C), (s), (iii) C) (A), (r), (ii) D) (D), (s), (iv)

15. Which of the following is INCORRECT combination?

A) (B), (r) , (i) B) (A), (P), (ii) C) (D), (s), (iv) D) (C), (q), (iv)

Answer Q,16, Q,17 and Q,18 by appropriately matching the information given in the three columns of the following table.

Column-I → contains the number N

Column-III → contains the corresponding values of Column-II

	Column I (N)	Column II	Column III
1)	$2^3 \cdot 3^5 \cdot 5^7 \cdot 7^9$	p) Number of even divisors of N	i) 24
2)	75600	q) Number of odd divisors of N	ii) 600
3)	$2^5 \times 3^4 \times 5^7 \times 7^2$	r) Number of divisors of N which are perfect squares	iii) 60
4)	720	s) Number of ways N can be resolved into two Factors	iv) 120

16. Which of the following is CORRECT COMBINATION.

A) (1), (P), (iii) B) (2), (s), (iii) C) (1), (s), (ii) D) (4), (q), (iii)

17. Which of the following is correct combination ?

A) (3), (p), (iii) B) (2), (r), (iv) C) (1), (r), (iv) D) (2), (q), (i)

18. Which of the following is INCORRECT combination?

A) (4), (p), (i) B) (2), (s), (iii) C) (3), (q), (iii) D) None of these

PHYSICS

Max. Marks: 61

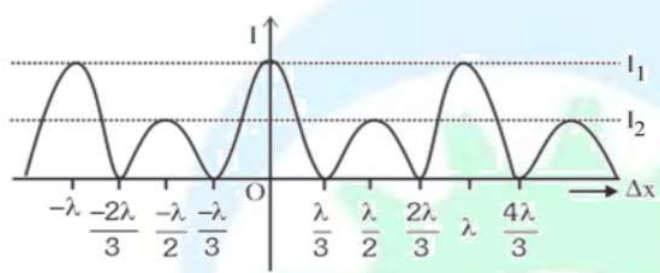
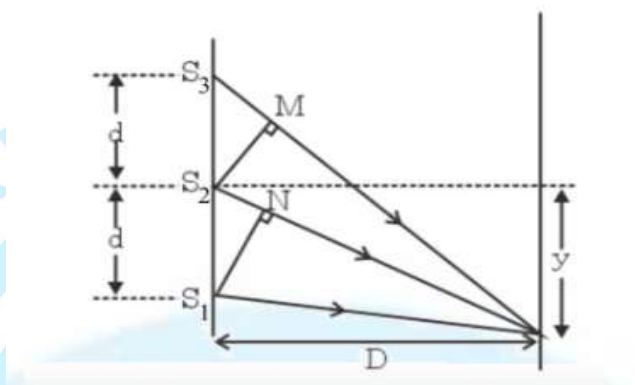
SECTION – I

(MULTIPLE CORRECT ANSWER TYPE)

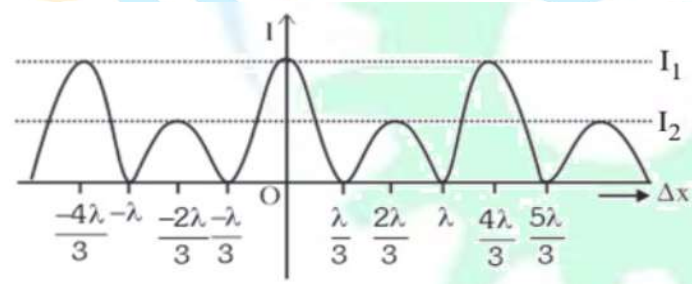
This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

19. A triple slit experiment is performed as shown in the figure with monochromatic light of amplitude A_0 . $S_3M = S_2N = \Delta x$. The graph between resultant intensity I versus Δx is plotted. Chose the correct option. ($d \ll D, y \ll D$)



A)

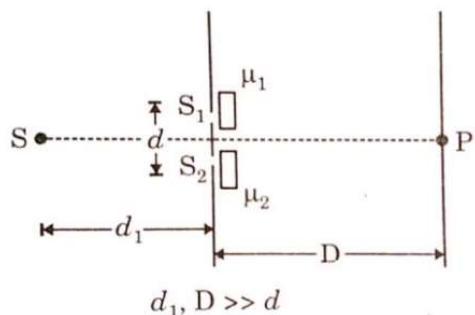


B)

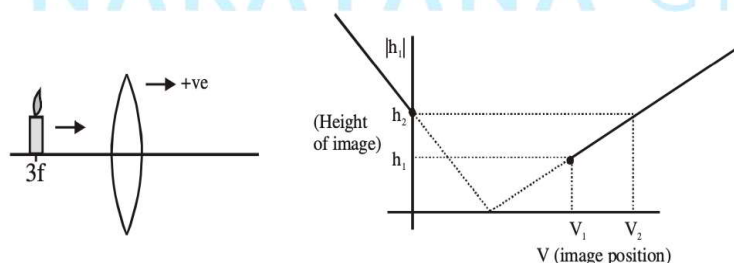
C) $\frac{I_1 + I_2}{I_1 - I_2} = \frac{5}{4}$, When I_1 and I_2 are the shown in above options.

D) $\frac{I_1 + I_2}{I_1 - I_2} = 5$, when I_1 and I_2 are the shown in above options.

20. In Young's experiment, the upper slit is covered by a thin glass plate of refractive index $\frac{4}{3}$ and of thickness 9λ , where λ is the wavelength of light used in the experiment. The other slit is also covered by another glass plate of thickness 2λ and refractive index $\frac{3}{2}$, as shown in figure. If I_0 is the intensity at point P due to slits S_1 and S_2 each, then:



- A) Intensity at point P is $4I_0$
- B) Two fringes have been shifted in upward direction after insertion of both the glass plates together
- C) Optical path difference between the waves from S_1 and S_2 at point P is 2λ
- D) If the source S is shifted upwards by a small distance d_2 then the fringe originally at P after inserting the plates, shift downward by $D\left(\frac{d_2}{d_1}\right)$
21. A candle of small height h is being moves along the axis from $u = 3f$ towards a thin converging lens of focal length f . The adjacent graph represents the height of image as function of the position of image w.r.t lens as shown in figure.



- A) The value of h_2 is h .
- B) The value of h_2 is $2h$.
- C) The value of h_1 is $h/2$
- D) The ratio v_1 / v_2 is $3/4$

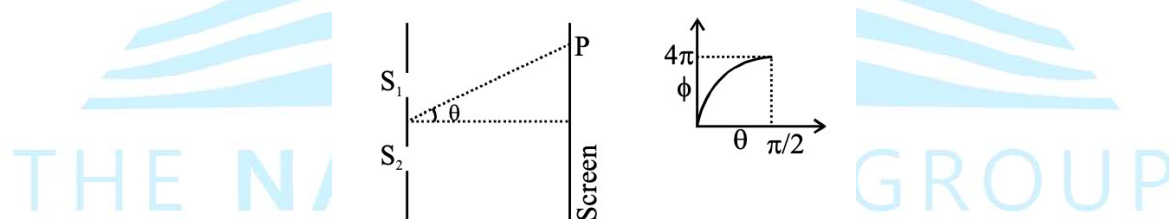
22. Four trials of Young's double slit experiment are conducted. (Given: $\lambda_{blue} = 450nm$ and $\lambda_{red} = 700nm$)

- i) In the first trial. Blue light passes through two fine slits $400\mu m$ apart and forms an interference pattern on a screen $4m$ away.
- ii) In a second trial, red light passes through the same slits and falls on the same screen.
- iii) A third trial is performed with red light and the same screen, but with slits $800\mu m$ apart
- iv) A final trials is performed with red light, slit $800\mu m$ apart, and a screen $8m$ away.

If angle between the first maxima and central maxima in each trial is $\theta_i, \theta_{ii}, \theta_{iii}$ & θ_{iv} respectively and the distance between the central maxima and the first maxima is $y_i, y_{ii}, y_{iii}, y_{iv}$ respectively, then:

- | | |
|--|--------------------------------------|
| A) $\theta_i > \theta_{ii} > \theta_{iii} > \theta_{iv}$ | B) $y_i > y_{ii} > y_{iii} > y_{iv}$ |
| C) $\theta_{ii} > \theta_i > \theta_{iii} = \theta_{iv}$ | D) $y_{ii} = y_{iv} > y_i > y_{iii}$ |

23. In a YDSE setup, we plot the phase difference (ϕ) between both waves at point P on the screen against the angular position (θ) of point P on the screen. The graph is as shown below.



- A) The distance $S_1S_2 = 2\lambda$
- B) There are total of 4 minima on the screen
- C) The first maxima above the centre is at $\theta = \frac{\pi}{4}$
- D) At $\theta = \frac{\pi}{6}$, intensity is maximum

24. Mark the CORRECT statements(s):

(Assume small aperture size for any lens or mirror involved in the bellow options)

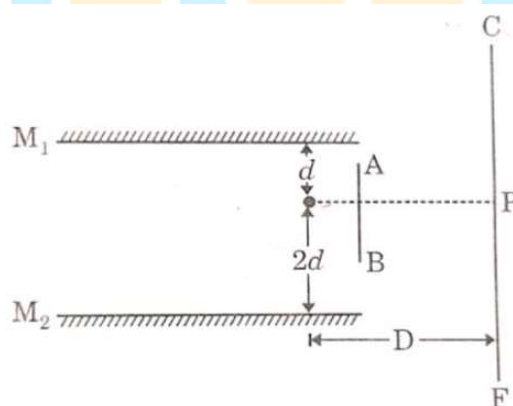
A) Direction of wave propagation is along the normal to wavefront.

B) For a point source of light, the shape of wavefront can be considered to be plane at very large distance from the source.

C) A point source of light is placed at the focus of a thin spherical lens, then the shape of the wavefront for emerged light may be plane.

D) The shape of wave wavefront for the light incident on a thin spherical lens (kept in vacuum) is plane, the shape of the wavefront corresponding to emergent light would be always spherical

25. M_1 and M_2 are two plane mirror and S is monochromatic source. AB is a 'opaque screen' which stops direct light to reach CF and allows reflected light from M_1 and M_2 to reach CF. P is a point on CF such that SP is parallel to M_1 and M_2 and perpendicular to CF:



A) Circular fringes will be formed on CF

B) P will be central maxima

C) P will be a point of maxima if $\lambda = \frac{6d^2}{D}$

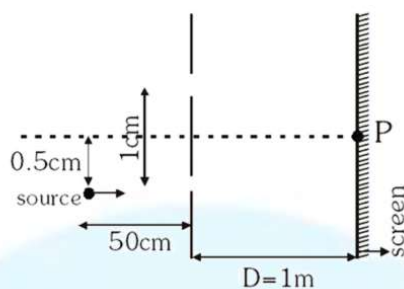
D) P will be a point of maxima if $\lambda = \frac{d^2}{D}$

SECTION-II
(INTEGER ANSWER TYPE)

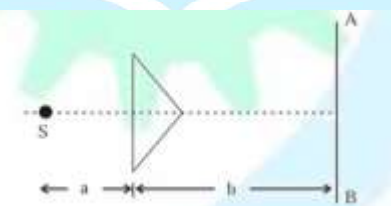
This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.

26. In a typical Young's double slit experiment a point source of monochromatic light is kept as shown in the figure. If the source is given an instantaneous velocity $v = 1 \text{ mm/s}$ towards the screen, then the instantaneous velocity of central maxima is given as $\alpha \times 10^{-\beta} \text{ m/s}$ upward in scientific notation. Find the value of $\alpha + \beta$.



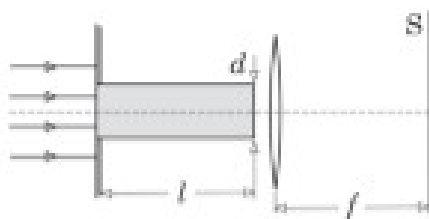
27. Light from source S is incident on the Fresnel biprism shown in figure. The light beams refracted by the different faces of the prism partly overlap and produce an interference pattern on a screen on its section AB. Find the distance between adjacent interference fringes if the distance from the source to the prism is $a = 1 \text{ m}$ and that from the prism to the screen is $b = 4 \text{ m}$. The angle of refraction of the prism is $\alpha = 2 \times 10^{-3} \text{ rad}$. The glass of prism has a refraction index of $n = 1.5$. The wavelength of the light is $\lambda = 6000 \text{ \AA}$. Express your answer in $\frac{x}{3} \times 10^{-2} \text{ cm}$.



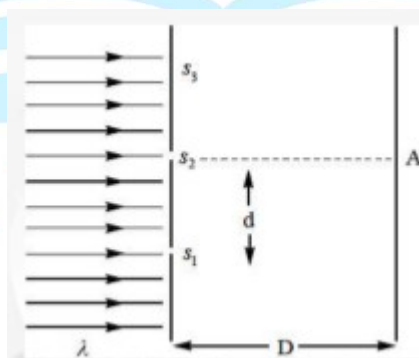
28. In the middle of a calm lake, a radio receiver is installed on a pole at a height 4.0 m above water level. The receiver is used to track radio signals from a satellite orbiting the earth. As the satellite rises above the horizon, intensity of the signals at the receiver varies periodically. It is minimum, when the satellite is 3° above the horizon and the again when the satellite is 6° above the horizon. Wavelength of the satellite signals is closest to $84/X$ (in cm). Find the nearest integer of X

29. A transparent rectangular slab of length $l = 20$ cm and thickness $d = 4.0$ cm is placed in front of a converging lens of focal length $f = 20$ cm. A screen S is placed in the focal plane of the lens. Refractive index of the material of the slab increases linearly from $\mu_0 = 1.5$ at the bottom face by an amount $\delta\mu = 2.0 \times 10^{-4}$ between the bottom and the top faces. A parallel beam is made to incident on the slab as shown in the figure.

The height of final image formed after refraction from optical axis is $X \times 10^{-2}$ cm ? find X



30. In standard YDSE ($D \gg d \gg \lambda$) with identical slits S_1 and S_2 light reaching a point A on the screen opposite to slit S_2 has an intensity I. It was also found that when only one of the two slits S_1 or S_2 was illuminated by same light beam, the intensity at A is still I. Now when a third slit S_3 of four times the slit width is made as shown ($S_1 S_2 = S_2 S_3 = d$) and all the three slits are illuminated then intensity of light reaching A is nI where $n =$ _____ (Assume intensity of source proportional to width)



SECTION – III

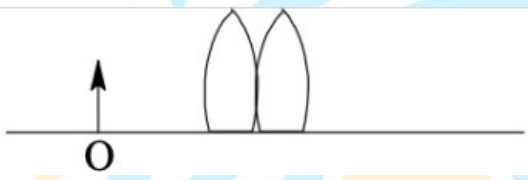
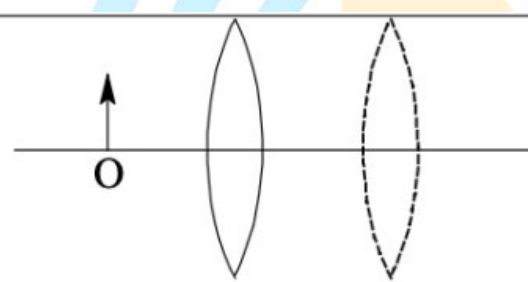
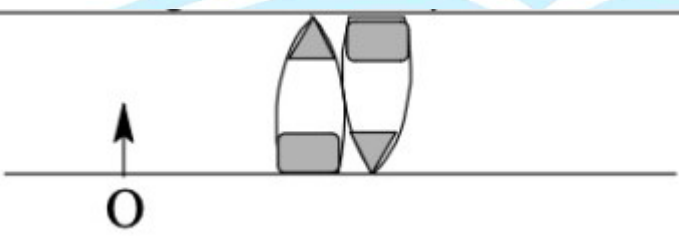
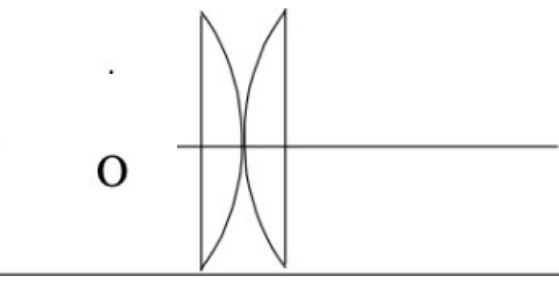
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q,31, Q,32 and Q,33 by appropriately matching the information given in the three columns of the following table.

A thin biconvex lens small aperture and of focal length f forms image of an object having certain intensity. If this lens is cut into two equal parts in two ways and are used to form image of the same object placed at same distance. In the table below column-1 represents certain ways in which lens or combination lenses is placed, column-2 represent the focal length of the lens or combination of lenses and column-3 represent the intensity of image in comparison to formed by complete lens:

Column-1	Column-2	Column-3
I) 	i) f	P) Decreases
II)  Same lens is shifted rightward to get two images at same position	ii) $f/2$	Q) Increases
III)  Shaded portion is painted black	iii) $2f$	R) Remain same
IV) 	iv) Infinity	S) Indetermine

31. Which combination is correct?

- A) (I) (iii) (Q) B) (II) (ii) (Q) C) (I) (ii) (P) D) (III) (iv) (R)

32. Which combination is used for calculating focal length of a convex lens?

- A) (IV) (ii) (Q) B) (II) (i) (P) C) (II) (i) (R) D) (III) (iii) (R)

33. Which combination is correct?

- A) (III) (iv) (S) B) (III) (ii) (P) C) (IV) (iv) (P) D) (IV) (i) (Q)

Answer Q,34, Q,35 and Q,36 by appropriately matching the information given in the three columns of the following table.

In a standard YDSE set up, a slab of thickness t & refractive index μ is placed in front of the upper slit. Other symbols have usual meaning.

Column-1		Column-2		Column-3	
I)	$\frac{D}{d} = 10^3$	i)	$t = 1 \mu m$ $\mu = 1.6$	P)	$\lambda = 400 \text{ nm}$
II)	$\frac{D}{d} = 2 \times 10^3$	ii)	$t = 1.5 \mu m$ $\mu = 1.3$	Q)	$\lambda = 500 \text{ nm}$
III)	$\frac{D}{d} = 5 \times 10^4$	iii)	$t = 2 \mu m$ $\mu = 1.6$	R)	$\lambda = 600 \text{ nm}$
IV)	$\frac{D}{d} = 4 \times 10^3$	iv)	$t = 2.5 \mu m$ $\mu = 1.4$	S)	$\lambda = 700 \text{ nm}$

34. In which of the following case, the central maxima shifts by 1mm?

- A) (II) (i) (P) B) (III) (ii) (R) C) (IV) (iii) (Q) D) (I) (iv) (S)

35. In which of the following case, the central maxima shifts by 2 fringe width?

- A) (I) (i) (P) B) (II) (ii) (Q) C) (III) (iii) (R) D) (IV) (iv) (S)

36. If the intensity at central maxima is I_0 , in which of the following case intensity at the centre of the screen is zero?

- A) (I) (i) (P) B) (II) (ii) (Q) C) (III) (iii) (R) D) (IV) (iv) (S)

CHEMISTRY**Max. Marks: 61****SECTION – I****(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

37. Which of the following property(s) is/are related to rosy red complex of Ni^{2+} and ammonical dimethyl glyoxime?
- A) Geometry around Ni is square planar and diamagnetic in nature
- B) Complex is stabilized by internal hydrogen bonding
- C) It is an organometallic complex
- D) Five membered chelate ring is formed when each dimethyl glyoximate is bonded with metal ion.
38. Consider $[\text{Ni}^{\text{II}}(\text{L})_n]^{\pm x}$ (where Ni = nickel, L = ligand and n = number of ligand) and select **CORRECT** observation for complex.
- A) If $\text{L} = \text{NH}_3$; $n = 6 \Rightarrow \mu = \sqrt{8}$ B.M.
- B) If $\text{L} = \text{Cl}^-$; $n = 4 \Rightarrow \mu = \sqrt{8}$ B.M.
- C) If $\text{L} = \text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$; $n = 3 \Rightarrow \mu = 0$
- D) If $\text{L} = ^-\text{CN}$; $n = 4 \Rightarrow \mu = 0$
39. Which of the following statements is /are **INCORRECT**?
- A) The oxidation state of iron in sodium nitroprusside $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NOS}]$ is +2
- B) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is tetrahedral in geometry
- C) In $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, Fe is d^2sp^3 hybridised
- D) In $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ complex ion, Co is sp^3d^2 hybridised

'X' $\xrightarrow[\text{Air}/\Delta]{\text{NaOH/KOH}}$ 'Y' $\xrightarrow{\text{O}_3}$ KMnO_4

Black Colour Compound Green Colour Compound (Molten)

'Y' $\xrightarrow{\text{H}_2\text{O/OH}^-}$ Green solution $\xrightarrow[\text{on evaporation}]{\text{NaOH}}$ 'Z'

D) 'Y' produce 'X' in the form of brown ppt. & KMnO_4 with water & CO_2

D) $[\text{Mn}(\text{H}_2\text{O})_6]^{+2}$: $\mu_{\text{observed}} > \mu_{\text{spin}}$; $[\text{Co}(\text{H}_2\text{O})_6]^{+3}$: Paramagnetic

D) due to equal contribution of entropy and enthalpy change

SECTION-II
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

44. The magnetic moment of the solid complex $[\text{Co}(\text{NH}_3)_6][\text{FeF}_6]$ to the nearest integer is (here Co and Fe are higher stable oxidation states)
45. The complex $[\text{Zn}(\text{acac})_2]^0$ has a spiro structure with Zn atom at the center. What will be the number of carbon atom(s) inside the spirocyclic arrangement of the molecule?
46. Consider the following complexes:
 $\text{K}_2[\text{CoF}_6]$; $\text{O}_2[\text{AsF}_6]$; $\text{K}_2[\text{CoCl}_4]$; $\text{K}_3[\text{Co}(\text{CN})_6]$; $\text{K}_3[\text{CoF}_6]$; $[\text{Ni}(\text{NH}_3)_6]\text{SO}_4$;
 $[\text{Pt}(\text{NH}_3)_6][\text{Fe}(\text{CN})_6]$;
 $\text{Na}_2[\text{Fe}^{\text{II}}(\text{CN})_5\text{NO}]$
 Determine the value of $[(Z-X)]$
 Given that: X = No. of paramagnetic complexes.
 Z = No. of complexes having diamagnetic cationic part.
47. Consider the following complexes:
 $[\text{Ni}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Cr}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{EDTA})]^-$,
 $[\text{CoF}_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
 Find $[X - Y]$
 X = No. of complexes having d^2sp^3 hybridisation
 Y = No. of complexes having sp^3d^2 hybridisation
48. The most acidic oxide of chromium on treatment with calculated amount of aqueous ammonia, followed by acidification with dilute HCl gives a compound 'A'. 1 mole of 'A' on heating produces 1 mole of a colourless diatomic gas 'B' and residue 'C'. The bond order in 'B' is X and oxidation number of Cr in residue 'C' is Y. Find the value of $(X + Y)$

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q,49, Q,50 and Q,51 by appropriately matching the information given in the three columns of the following table.

COLUMN – I			COLUMN – II		COLUMN – III
I)	Eu	i)	$4f^7 5d^1 6s^2$	P)	Hardest Lanthonoid metal
II)	Gd	ii)	$4f^6 6s^2$	Q)	Melts at 1623 K
III)	Sm	iii)	$4f^7 6s^2$	R)	Highest number of unpaired e^- in M^{+3}
IV)	Lu	iv)	$4f^{14} 5d^1 6s^2$	S)	Atomic size is highest in ground state

49. Which of the following are **CORRECT** combination?

- A) (I) (i) (R) B) (III) (i) (R) C) (I) (ii) (S) D) (III) (ii) (Q)

50. Which of the following are **CORRECT** combination?

- A) (IV) (iv) (S) B) (IV) (iv) (P) C) (II) (iii) (R) D) (II) (iv) (S)

51. Which of the following are **CORRECT** combination?

- A) (IV) (iv) (Q) B) (III) (i) (R) C) (II) (ii) (S) D) (I) (iii) (S)

Answer Q,52, Q,53 and Q,54 by appropriately matching the information given in the three columns of the following table.

COLUMN – I			COLUMN – II		COLUMN – III
I)	$\text{PdCl}_2 \cdot 4\text{NH}_3$	i)	Has sp^3 hybridisation	P)	Has more than two planes of symmetry
II)	$\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$	ii)	Outer orbital complex	Q)	Dipole moment ≥ 0
III)	$\text{CoCl}_3 \cdot 4\text{NH}_3$	iii)	Has $d^2\text{sp}^3/\text{sp}^3d^2$ hybridisation	R)	2 Moles of AgCl precipitated per mole of the compounds with excess AgNO_3
IV)	$\text{PtCl}_4 \cdot 2\text{HCl}$	iv)	Inner orbital complex	S)	1 Moles of AgCl precipitated per mole of the compounds with excess AgNO_3

52. Select the **INCORRECT** combination from the following.

A) (I) $\rightarrow$ (R), (P), (Q)

B) (II) $\rightarrow$ (ii), (Q), (R)

C) (III) $\rightarrow$ (iii), (Q), (S)

D) (IV) $\rightarrow$ (iv), (Q), (S)

53. Select the **CORRECT** combination from the following.

A) (I) $\rightarrow$ (ii), (Q), (R)

B) (II) $\rightarrow$ (iv), (P), (R)

C) (III) $\rightarrow$ (iii), (Q), (R)

D) (IV) $\rightarrow$ (iv), (P), (Q)

54. Select the **INCORRECT** combination from the following

A) (I) $\rightarrow$ (iii), (Q)

B) (II) $\rightarrow$ (iii), (Q)

C) (III) $\rightarrow$ (iii), (S)

D) (IV) $\rightarrow$ (iii), (Q)

